

Série TP 1 : File Integrity Monitoring (FIM)

Practical Exercise: Building a File Integrity Monitoring (FIM) System

Objective:

Create a simple File Integrity Monitoring (FIM) tool in Python to:

1. Generate a baseline of file hashes for a directory.
2. Monitor the directory for additions, deletions, and modifications.
3. Alert the user when changes are detected.

Tasks:

4. Baseline Creation:

- Write a Python script to generate a baseline of file hashes (SHA-256) for a directory.
- Store file paths and their hashes in a dictionary.

5. Directory Monitoring:

- Compare the current state of the directory to the baseline to detect:
 - New files.
 - Deleted files.
 - Modified files.
- Display alerts when changes are detected.

6. Testing the System:

- Create a directory with test files.
- Generate the baseline.
- Modify the directory (add, delete, or modify files).
- Verify that the system detects changes.